MIL-A-87022B(NU)
16 October 1987
SUPERSEDING
MIL-A-87022A(NU)
6 May 1982

MILITARY SPECIFICATION

APRON, UTILITY, DISPOSABLE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a disposable utility apron fabricated from coated, spunbonded olefin fabric. The apron shall be of one type and one size.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Material Handling, Wood, Stringer Construction, 2 Way And 4 Way (Partial)
DDD-L-20	- Label; for Clothing, Equipage, and Tentage (General Use)

MILITARY

MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Box, Shipping Fiberboard (Modular Sizes)
MIL-C-87019	- Cloth, Nonwoven, Disposable, Spunbonded Olefin (Coated and Uncoated)

STANDARDS

FEDERAL

FED-STD-751	- Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

3.3 Material.

3.3.1 Basic material. The basic material for the apron including the tie tapes, shall be of a white polyethylene coated spunbonded olefin material conforming to type II, Class B of MIL-C-87019. As an alternate, the tie tapes may be made of an uncoated white spunbonded olefin material conforming to type II, Class A of MIL-C-87019.

3.3.2 Thread. The sewing thread shall be of any commercial type, suitable for the purpose intended.

3.3.3 Label. Each apron shall have a combination identification and cautionary statement label and shall conform to type IV or V, Class 15 of DDD-L-20. The label will not be less than 2 by 3 inches bearing the inscription specified below:

APRON, DISPOSABLE
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:

CAUTION

THIS GARMENT SHALL NOT BE REUSED AND
MUST BE DISCARDED WHEN OTTO FUEL
STAINS ARE OBSERVED.

The cautionary statement shall be printed in larger letters than all other lettering on the label.

3.4 Design. The apron shall have a high bib front with a scoop neck. Tie tapes shall be attached along both sides of the apron, which will allow adjustment at neck and rear. (See Figure 1).

3.4.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.5 Pattern. The standard pattern to be used to cut the working pattern will be furnished by the Government (see 6.3). The working pattern shall be identical to the Government pattern. Neither the Government pattern nor the working pattern shall be altered in any way.

3.5.1 Pattern parts. The apron shall be cut from the basic material specified in 3.3.1 and in accordance with the following pattern part:

MATERIAL	NOMENCLATURE OF PATTERN PART	CUT PARTS
Cloth, Nonwoven coated	Apron w/ bib	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

3.6 Construction.

3.6.1 Stitch, seam, and stitching types. The stitch, seam, and stitching types specified herein shall conform to FED-STD-751.

* 3.6.2 Thread breaks and ends of seams. Skipped stitches and thread breaks shall be repaired by stitching not less than 1/2 inch beyond each end of the defect using a 301 stitch. Ends of 301 stitching shall be backstitched, or overstitched, 1/2 inch minimum. Thread tension shall be maintained so there will be no loose stitching, or tight stitching which could cause breaks in the line of stitching.

3.6.3 Stitches per inch. The minimum and maximun number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operation requirements. The apron shall be manufactured in accordance with the operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations shown provided the finished aprons are identical to those produced by following the sequence as specified in Table I.

3.7.1 Abbreviations in table of operations. The abbreviations used in table I are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

TABLE I - CONSTRUCTION OF APRON

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
1.	Cutting. a. Cut the apron portion from the basic material specified in 3.3.1, in strict accordance with the pattern furnished which shows size, shape, and directional lines for cutting. b. Cut the two (2) tie tapes from the basic material specified in 3.3.1, aproximately 1 1/2 inches in width unfinished. The unfinished length of the tie tapes shall be cut in order to comply with the requirements specified in operation 3 c. c. The coated side of the material shall be the face and finish on the outside of the garment.					
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production.					
3.	Attach tie tapes. a. Stitch tie tapes by simultaneously inserting and catching each side of the bib portion. b. Each tie tape shall finish approximately 3/8 inch in width. c. The free ends of the tie tapes shall finish even in length, each free end not less than 30 inches long at rear and not less than 11 inches long at the neck. 1/ For thread type refer to 3.3.2.	301	BSc-1 and EFp-1	6-8	1/	1/
4.	Mark or attach label. Mark or adhere the label to the inside of the apron approximately 2 inches below the neckline and centered.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

3.8 Measurements. The finished apron shall conform to the measurements as shown in Table II. All measurements and tolerances are expressed in inches, and shall be made with the apron placed smooth and flat.

Table II - Finished Measurements

Measurement	Dimensions	Tolerance
Overall length (A)	45 1/2	± 1
Width at top, excluding bib (B)	28	± 1
Bottom width (C)	28	± 1

NOTE: A thru C refer to Figure 1

A. Measurement shall be taken from the center high point of scoop neck to bottom edge of apron.
B. Measurement shall be taken across the front from side to side at intersection of sides and start of bib.
C. Measurement shall be taken across the bottom from side to side.

3.9 Workmanship. The finished aprons shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2 and 6.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The sample unit shall be one apron. The lot size shall be expressed in units of aprons.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

4.4.2.1 Visual exmaination. The aprons shall be examined for defects classified in accordance with the list below:

Examine	Defect
Material Defects and Damages	Hole, cut, tear, needlechew, visible mend, or burn. Any evidence of delamination of the coating on outside of apron. Spots or stains, clearly visible at an inspection distance of 3 feet.
Seams and Stitching	Any open seam. NOTE: The seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur. Not specified seam or stitch type. Loose stitch tension resulting in a loose seam. Tight stitch tension (stitches break when normal strain is applied in the direction of the seam or stitching). Thread breaks not repaired or stitched less than 1/2 inch beyond each end of break. Stitches per inch more or less than required. Ends of stitch lines not backstitched/overstitched as required.
Components and Assembly	Any component part or operation omitted. Any component part misplaced or operation improperly performed. Any component part not securely affixed, e.g., tie tape loosely attached to length side of bib. Coated side of cloth on inside of garments.
Label	Missing, incorrect, illegible, or misplaced. Cautionary statement lettering not larger than all other lettering. Not size specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

4.4.2.2 Dimensional examination. The apron shall be examined for defects in dimensions in accordance with the following:

a. Any measurement deviating from nominal dimensions and tolerances specified in Table II, shall be scored as a defect.
b. Finished free ends of the tie tapes unequal in length by 1 inch or more, shall be scored as a defect.
c. Finished free ends of tie tapes at neck less than 11 inches in length, shall be scored as a defect.
d. Finished free ends of tie tapes at rear less than 30 inches in length, shall be scored as a defect.

4.4.2.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and acceptable quality levels (AQL's), expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection levels	AQL
For defects applicable to 4.4.2.1	II	2.5
For defects applicable to 4.4.2.2	S-3	4.0

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width,or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level A or C as specified (See 6.2).

5.1.1 Level A. Ten (10) aprons, placed one on top of the other shall be folded to measure approximately 14 1/2 by 11 1/2 inches.

5.1.2 Level C. The aprons shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be tha same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Two hundred and forty (240) aprons, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to type CF, variety DW, grade V15c, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and twelve in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

5.2.2. Level B. Two hundred and forty (240) aprons, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety DW, grade 275, size 3A, of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and twelve in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Item packaged, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII, fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The apron is intended primarily for use as a part of a protective ensemble used to protect torpedo handlers from exposure to Otto fuels.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Whether first article sample is required (see 3.2 and 4.3), the item shall be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, quantity, and testing and approval of the first article.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When palletization is required (see 3.1).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Subject term (key word) listing.

Apron
Polyethylene coated
Clothing, disposable
Nonwoven, spunbonded
Olefin
Otto fuel protective

Custodian:
Navy - NU

Review Activity:
Navy - OS
DLA - CT

User Activity:
Navy - SH

Preparing Activity:
Navy - NU

Project No. 8415-N569

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

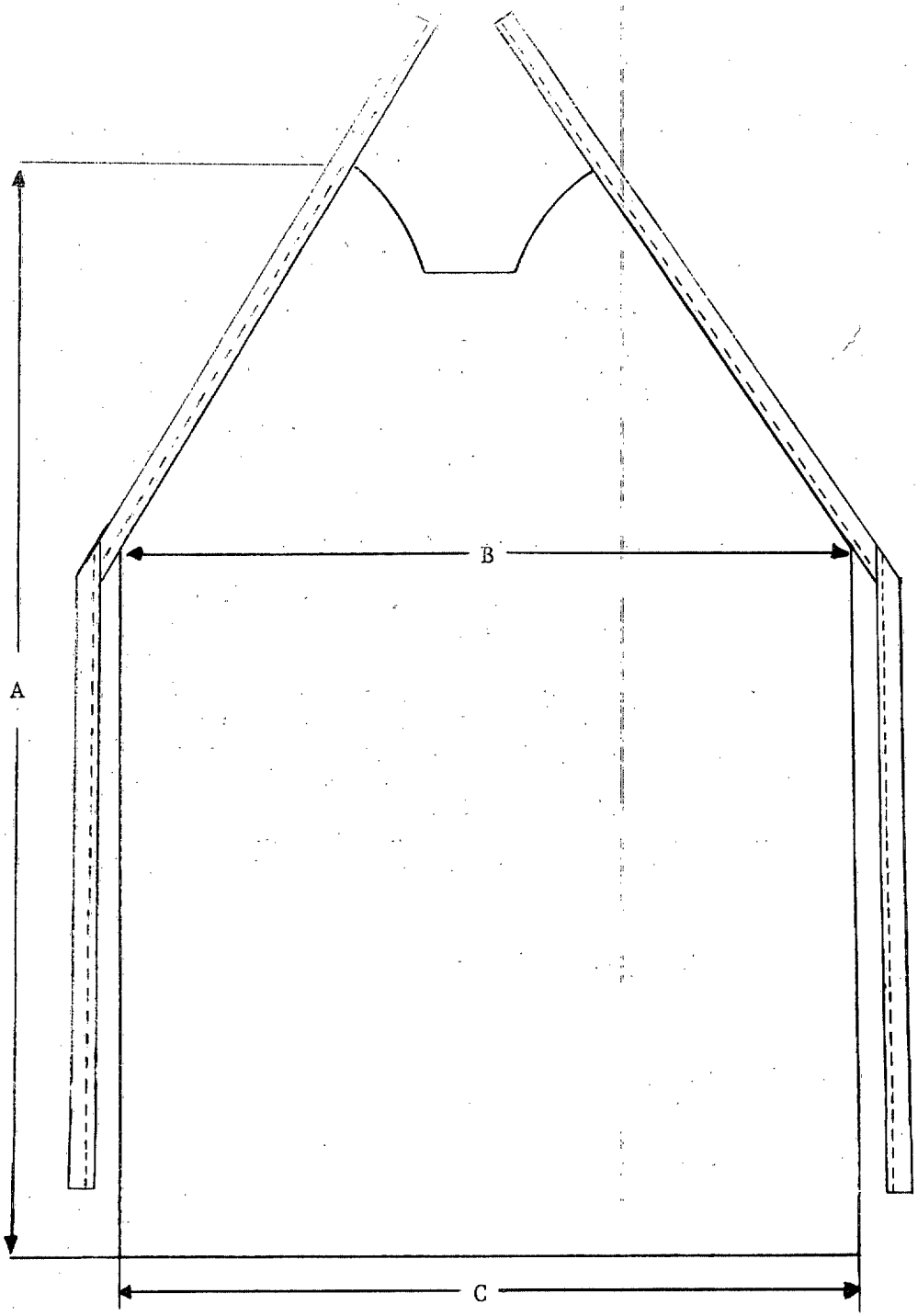


FIGURE 1 - APRON, UTILITY, DISPOSABLE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-87022B(NU)	APRON, UTILITY, DISPOSABLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Officer in Charge
Navy Clothing & Textile Research Facility
21 Strathmore Road (Code 50)
Natick, MA 01760-2490

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:35Z
Check the source to verify that this is the current version before use.